

Multimodal Evidential Learning for Open-World Weakly-Supervised Video Anomaly Detection

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Abstract—Efforts in weakly-supervised video anomaly detection center on detecting abnormal events within videos by coarse-grained labels, which has been successfully applied to many real-world applications. However, a significant limitation of most existing methods is that they are only effective for specific objects in specific scenarios, which makes them prone to misclassification or omission when confronted with previously unseen anomalies. Relative to conventional anomaly detection tasks, Open-world Weakly-supervised Video Anomaly Detection (OWVAD) poses greater challenges due to the absence of labels and fine-grained annotations for unknown anomalies. To address the above problem, we propose a multi-scale evidential vision-language model to achieve open-world video anomaly detection. Specifically, we leverage generalized visual-language associations derived from CLIP to harness the full potential of large pre-trained models in addressing the OWVAD task. Subsequently, we integrate a multi-scale temporal modeling module with a multimodal evidence collector to achieve precise frame-level detection of both seen and unseen anomalies. Extensive experiments on two widely-utilized benchmarks have conclusively validated the effectiveness of our method. The code will be made publicly available.

Index Terms—Video anomaly detection, vision-language model, evidential learning.

I. INTRODUCTION

VIDEO anomaly detection (VAD), which aims to identify the anomaly events that diverge from normal ones, has become an increasingly critical field because of its promising

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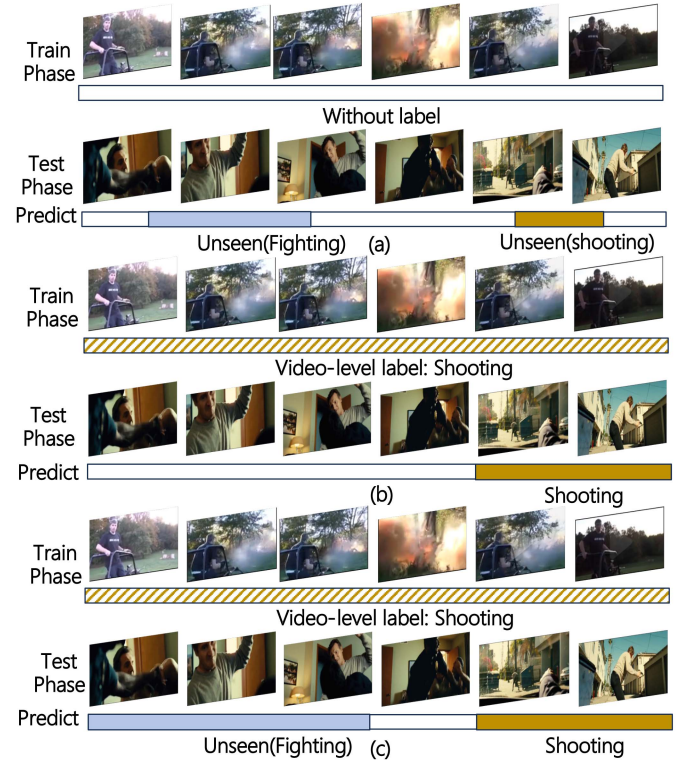


Fig. 1. Illustration of the train and test phases of (a) Unsupervised, (b) Weakly-supervised and (c) Open-world Weakly-supervised VAD task.

application prospects. Due to the unavailability of substantial fine-grained anomaly annotation, most existing VAD methods focus on unsupervised learning and weakly-supervised learning to detect those deviated samples as anomalies. The main difference between them is inherent in the availability of annotated training samples. As shown in Fig. 1(a) and (b), unsupervised VAD methods [1], [2], [3], [4] focus on how to detect unseen anomaly events by mining the intrinsic structural information and correlation of data, while Weakly-supervised VAD (WVAD) methods [5], [6], [7], [8], [9], [10] focus on how to detect seen anomaly events by limited annotated training samples. From the perspective of supervised mode and model performance, neither method is entirely satisfactory for real-world VAD. On the one hand, though unsupervised methods are convenient for detecting a wide variety of unseen anomaly events in the real world,

the performance is often unsatisfactory. On the other hand, while WVAD methods exhibit superior performance in closed-set scenarios, they manifest a susceptibility to misclassification or oversight when confronted with unanticipated anomaly events. Some recent works [11], [12], [13], [14], [15], [16] have attempted to solve this problem by means of image enhancement, which improves its performance in the real world to some extent, but still does not completely solve this problem.

The essential reason for this situation is that, as the dynamically changing real world continues to evolve, never-before-seen anomalies consistently emerge. Therefore, to meet the requirements of VAD tasks in the real world, it is necessary to explore the Open-world WVAD (OWVAD) methods, which can achieve content performance in detecting anomaly events from videos with assorted seen and unseen anomaly.

Divergent from the traditional VAD methods, as shown in Fig. 1(c), OWVAD methods can detect temporal boundaries accurately of the anomaly events in the testing phase, which have never occurred in the training phase. There is no doubt that OWVAD tasks are more complicated than WVAD tasks based on closed sets. We argue that the following two aspects are key limiting factors for OWVAD performance:

- *Complexity of scenarios and diversity of anomaly categories in the real world:* Unlike the closed-set WVAD task, the OWVAD task is expected to detect anomaly events from a variety of real-world scenarios. However, the limited scenario information in the training set makes it difficult for the model to have a generic visual representation. Moreover, a single scenario usually contains a series of associated anomaly events, which is difficult to detect accurately in the absence of information about the internal correlation of these associated events.
- *Ambiguity of training sample annotations:* Under the conditions of OWVAD, not only are annotations unavailable for the unseen anomaly, but also the temporal boundaries for both seen and unseen anomaly can only be ambiguously inferred from the video and its coarse-grained annotation. This ambiguity brings confusion between seen anomaly, unseen anomaly, and background. Furthermore, the duration of anomalous events is often unpredictable and exhibits considerable variability in the real world, which increases the difficulty of accurately identifying their temporal boundaries.

To alleviate the above limitations of OWVAD, we propose a Multi-scale Evidential Learning method based on the Vision-Language Pre-trained model (MEL-VLP) for OWVAD. Guided by the introduced visual-textual associated knowledge, MEL-VLP detects both seen and unseen anomalies by collecting multi-scale temporal contexts, text-associated information, and joint-modal features as evidence. To be specific, we first extract visual features and textual features from the encoders of vision-language pre-trained model and constructed a joint discriminative framework to detect anomalies by both visual-textual features and their associated relationship. Recently, large vision-language pre-trained model has exhibited remarkable success across a diverse array of downstream tasks, attributed to its acquired cross-modal prior knowledge and robust

transfer learning capabilities. Some recent VLP-based WVAD methods [17], [18] have demonstrated the feasibility and superiority of VLP (e.g. Contrastive Language-Image Pre-training (CLIP [19])) in VAD task.

Second, to address the problem of ambiguity of training sample annotations, we design a multimodal evidence collector, under Subjective Logic Theory (SLT [20]), to better distinguish between background and anomalies by jointly leveraging single-modal and multimodal features. We argue that, for abnormal events in a scenario, visual features should provide sufficient evidence of the seen anomalies, while textual features can provide not only supplementary evidence information for seen anomalies but also correlation evidence for unseen anomalies, thereby enhancing the model's ability to detect various anomalies in the real world. To combine multimodal evidence to dynamically calibrate the temporal boundaries of anomaly, we propose a multimodal evidential collaborative learning method that allows single and joint modal evidence information to be calibrated against each other according to their respective confidence and uncertainty.

Finally, for the unpredictability and variability of anomaly duration, a Multi-scale Temporal Visual Modeling (MTVM) module is designed to empower single snippet features with the capacity to perceive multi-scale contextual video segments. By capturing and aggregating local and global visual information across video segments that traverse distinct temporal scales, our MTVM facilitates the acquisition of features at both fine-grained and coarse-grained levels, thereby augmenting the capability to characterize anomalies with varied durations.

Therefore, our potential contributions can be summarized as follows.

- We propose a novel multi-scale evidential learning framework for open-world weakly-supervised video anomaly detection, which is driven by vision-language pre-trained model and under the guidance of subjective logic theory.
- We jointly leverage the multi-scale temporal visual modeling (MTVM) module and multimodal evidential collaborative learning (MECL) module to address the limitation of ambiguity of training sample annotations.
- We conduct extensive experiments on two popular benchmarks, XD-Violence and UCF-Crime. And the experimental results demonstrate significant superiority to several state-of-the-art weakly supervised approaches.

II. RELATIVE WORK

A. Weakly Supervised Video Anomaly Detection

Recent weakly supervised anomaly detection methods are mainly based on the multi-instance learning (MIL) framework [21], which regards the anomaly and normal segments as positive and negative samples respectively. Among them, most of the methods can be classified into three categories, namely, erasing-based methods [22], [23], [24], [25], attention-based methods [17], [26], [27], [28], [29] and uncertainty-based methods [30], [31], [32]. Erasure-based methods mainly erase outstanding areas by setting a threshold and then mining less obvious areas. For example, DE-Net [22] devises a dynamic erasing

strategy that dynamically evaluates the completeness of the detected anomalies and erases the prominent anomaly segments in order to motivate the model to discover milder anomaly segments in the video. Attention-based methods suppress background components by assessing foreground attention scores mainly through the attention mechanism. RTFM [33] employs self-attention mechanisms and dilation convolutions to capture both long-range and short-range temporal dependencies for better feature learning.

As an emerging sort of approach, uncertainty-based methods primarily decompose model classification probabilities into confidence and uncertainty through uncertainty estimation, and optimization is conducted based on this uncertainty. Although this paradigm has been successful in many fields [34], [35], it has only been little studied in the field of anomaly detection. For example, CELL [36] proposes a Cascade Evidential Learning framework at an evidence level, which jointly leverages multi-scale temporal contexts and knowledge-guided prototype information to achieve temporal action localization. For video anomaly detection, Zhu et al. [30] use evidential deep learning to instantiate MIL and use the predicted evidence to help select anomaly instances with high cleanness for robust MIL training. In this paper, by collecting multimodal anomaly evidence, we propose a novel multimodal evidence collaboration framework driven by vision-language pre-trained model to effectively mitigate the background ambiguity problem in WVAD.

B. Vision-Language Pre-Training Model

Over the past few years, vision-language pre-trained models [37], [38], [39], [40] have achieved substantial success, attributed to their pre-training on large-scale visual-textual pairs. Pre-training based on large amounts of data allows the model to learn more general and efficient visual and textual feature representations. Of greater significance is the assistance rendered by the inherent associative information between visual and textual modalities learned during the pre-training process, facilitating application to diverse multi-modal downstream tasks [41], [42], [43]. Llama [44] extends the scope of modeling in multimodal domains such as image and video by integrating visual inputs with the capabilities of LLMs. DoraemonGPT [38] achieves better processing of video tasks by combining LLM with dynamic scene understanding to reason and predict objects and scenes in videos.

For the VAD task, some recent works achieved significant performance gains by introducing the encoders of vision-language pre-trained models. For instance, CLIP-TSA [17] acquires more discriminative visual features using visual coders from CLIP and models temporal dependence using a self-attention mechanism. Nevertheless, existing works have not delved deeply into the exploration of associative information from vision-language pre-training model for the OWVAD task. In this paper, we introduce visual encoders and textual encoders from vision-language pre-train model to learn more generalized feature representations. At the same time, we enhance the performance of the model in the open world by learning the intrinsic correlation information between visual and textual modalities.

C. Evidential Deep Learning

Grounded in subjective logic theory [20], evidence-based deep learning involves the collection of scalar evidence for each category and models the probability distribution of the collected evidence, enabling uncertainty estimation in a single forward pass. Specifically, subjective logic is a logical framework for dealing with and reasoning about uncertainty and trust, which is particularly well suited for modeling and analyzing subjective opinions or beliefs. It combines ideas from probabilistic and multi-valued logic to be able to deal with the complexity of uncertainty, incomplete information, and inter-subjective trust. In contrast to traditional probabilistic logic, subjective logic theory formalizes the cognitive uncertainty of a model in the face of incomplete information by assigning belief measures to potential categories to characterize assertions of truthfulness and explicitly incorporating uncertainty measures. As an emerging method for uncertainty assessment, it holds a significant position across numerous applications. For video anomaly detection, Zhu et al. [30] use evidential deep learning to instantiate MIL and use the predicted evidence to help select anomaly instances with high cleanness for robust MIL training. Huang et al. [45] propose a model called Uncertainty-Aware Prototypical Transformer (UPformer), which is specialized for visual anomaly detection at the pixel level. In this work, the authors specifically emphasize two challenges in the anomaly detection task: the diversity of anomalies and the blurred boundaries between them and normal regions. To address these challenges, the authors design a novel approach that takes into account not only the diversity of anomalies but also the uncertainty in the anomaly detection process.

However, evidence collected under single-modality conditions is limited by the constraints of modality-specific information. Current methods used for VAD tasks overlook the correlation and collaboration of evidence in a multi-modal context. Therefore, we propose a multi-modal evidence collaboration method aimed at endowing the model with the capability to dynamically calibrate anomaly temporal boundaries.

III. METHOD

A. Overall Architecture

Problem Definition: OWVAD aims to detect seen and unseen abnormal instances in videos. Given a set of weakly-labeled training videos $V = \{\mathcal{X}^{(k)}, y^{(k)}\}_{k=1}^{|V|}$, where $\mathcal{X}^{(k)} \in \mathbb{R}^{N_k \times H \times W}$ represents a video with N_k frames with $H \times W$ pixels, and $y^{(k)} \in \mathbb{R}^C$ is its multi-hot label indicating the anomaly categories that the abnormal instances in this video belong to, where C is the number of seen anomaly categories. During the testing phase, the model is used to predict frame-level anomaly scores for the test video. It is worth noting that in the OWVAD task, the training data contains several unknown anomaly categories, which are labeled as unknown categories.

Overview: We firstly leverage the most popular visual-language pre-trained models, such as CLIP, utilizing their visual encoders and text encoders to extract visual and textual features, which possess greater generality and discriminability. As shown

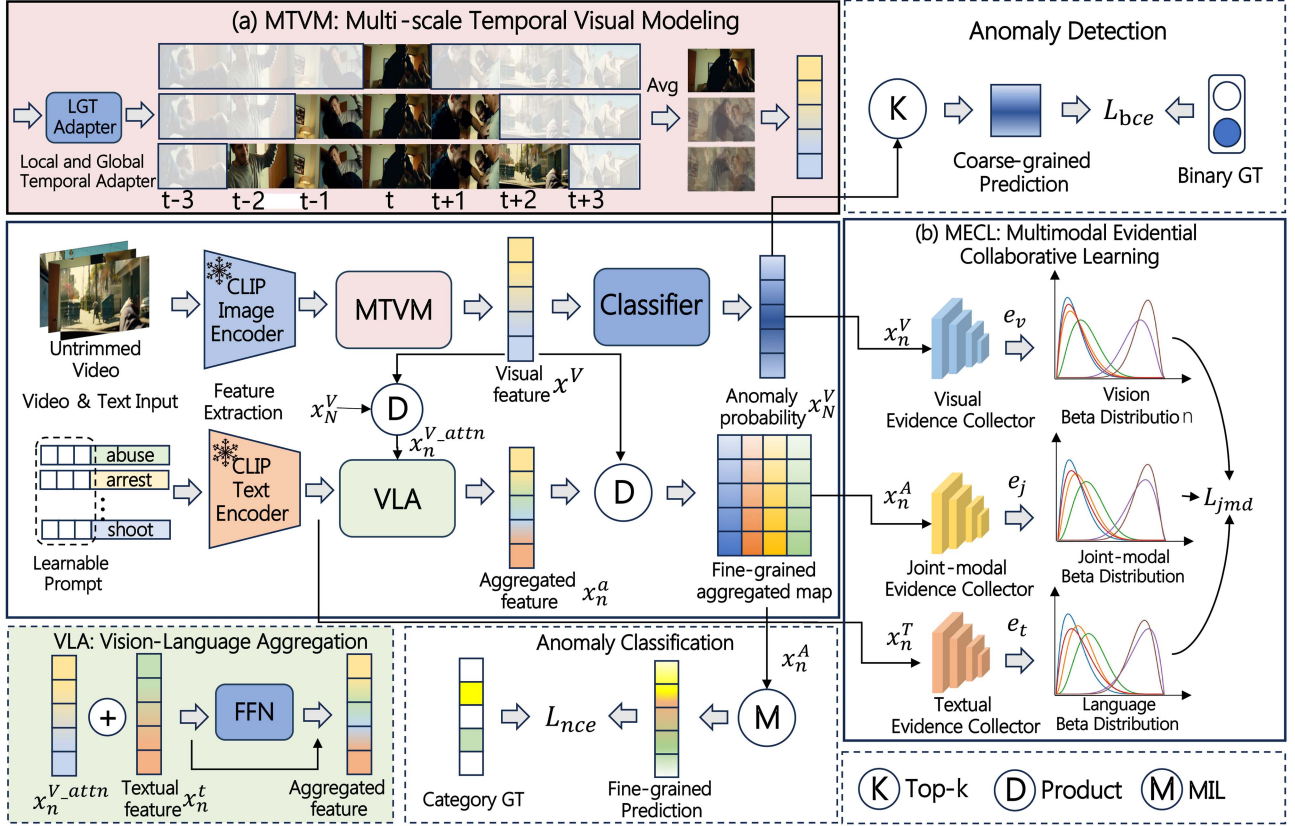


Fig. 2. The whole framework of our proposed method, in which the solid line box is divided into the main novel part of this paper. First, we perform feature extraction on the video and labeled text with CLIP’s image encoders and text encoders to learn the more general and discriminative features. We then propose the (a) Multi-scale Temporal Visual Modeling (MTVM) module for sensing different ranges of temporal contexts and capturing useful information. Finally, we design the (b) Multimodal Evidence Collaborative Learning (MECL) module to dynamically calibrate the temporal boundaries of anomaly events.

in Fig. 2, for a given training video V , we input it into the visual encoders of CLIP to obtain visual features $X^V \in \mathbb{R}^{N \times D}$, where N is the number of video frames, D is the dimension of visual feature. Similarly, we input category labels into the text encoders to obtain textual features $X^T \in \mathbb{R}^{(C+2) \times D}$, where C is the number of anomaly classes. Specifically, we code ‘normal’ and anomaly categories as text here. On the one hand, visual features X^V employ the multi-scale temporal visual modeling (MTVM) module to perceive temporal context and derive anomaly probability $X_N^V \in \mathbb{R}^{1 \times N}$ through the classifier. Subsequently, we can get frame-level anomaly scores through the Tok-k algorithm. On the other hand, visual attention $X_{N,N}^{V_attn} \in \mathbb{R}^{1 \times D}$ (the production between visual features X_N^V and anomaly probability X_N^V) and textual features are aggregated together through the Visual Language Aggregation (VLA) module to get aggregated features $x_n^a \in \mathbb{R}^{N \times D}$, and then a fine-grained aggregation mapping $x_n^A \in \mathbb{R}^{N \times (C+2)}$ is obtained through the product with the visual features.

Besides, we use the multimodal evidential collaborative learning (MECL) module to dynamically calibrate the temporal boundaries of anomaly. Specifically, we collect evidence from anomaly probability X_N^V , textual features X^T and aggregation mapping x_n^A , respectively. The corresponding Dirichlet distributions are obtained from the collected evidence and the learned evidence is dynamically corrected by a joint-modal

discrimination (JMD) strategy. Specifically, the training process and inference process are shown in Algorithms 1 and 2.

B. Multi-Scale Temporal Visual Modeling

Despite the potent feature representation capabilities of CLIP, it is trained on a large-scale dataset of image-text pairs, which implies that it faces challenges in extracting temporal dependencies within video data. Nevertheless, temporal dependencies play a crucial role in video anomaly detection. Therefore, inspired by [46], [47], [48], we employ a Local and Global Temporal (LGT) adapter to model the long and short-term temporal dependencies in videos. Then we design a multi-scale temporal visual modeling (MTVM) module to empower single snippet features with the capacity to perceive multi-scale contextual video segments.

Local and Global Temporal Adapter: To capture the local temporal dependencies, we first compute the internal self-attention of each video segment by Transformer. Then to capture the global temporal dependency, we model the global temporal dependency in terms of feature similarity and relative distance using GCN. These two processes can be expressed by the following formulas:

$$X_l = [Transformer(x_{seg1}), \dots, Transformer(x_{segj})], \quad (1)$$

$$X_g = \text{gelu}([\text{Softmax}(H_{\text{sim}}); \text{Softmax}(H_{\text{dis}})]X_l W), \quad (2)$$

where segi and segj denote the serial numbers of different video segments respectively. X_l is the frame-level video feature extracted from the local module. H_{sim} is the adjacency matrix computed by cosine similarity of each frame. H_{dis} is the adjacency matrix computed by the relative positions between each two frames.

Multi-scale Temporal Visual Modeling: In MTVM, each frame is assigned a series of multiscale video segments centered around it, and the contextual awareness properties of these video segments contribute to the expansion of perceptual scope. When a series of video segments centered around t are given, our process of averaging this series of multiscale video frames can be expressed as follows:

$$f_n^m = \frac{1}{e_n^m - s_n^m + 1} \sum_{s_n^m \leq i \leq e_n^m} x_i, \quad (3)$$

where m is the length of the predetermined temporal range, and (s_n^m, e_n^m) indicate the start time and end time of the video segment.

To enhance the perceptual capabilities of frame n regarding abnormal events within the temporal context, feature enhancement is conducted through the fusion of x_n and f_n^m . We argue that frame-level visual features can assimilate informative contextual details from extended video segments highly correlated with them. Therefore, the fusion of x_n and f_n^m to obtain frame-level features with context awareness can be expressed as:

$$\tilde{x}_n = (1 - \beta) x_n + \beta \sum_{m \in \mathcal{M}} \omega_n^m f_n^m, \quad (4)$$

where $\mathcal{M}$ is the total number of fused fragments, β is a hyperparameter for fusion weight of the cosine similarities ω_n^m between x_n and f_n^m . And ω_n^m can be calculated by:

$$\omega_n^m = \frac{x_n f_n^m}{\|x_n\|_2 \cdot \|f_n^m\|_2}. \quad (5)$$

C. Multimodal Evidential Collaborative Learning

We design a novel multimodal evidential collaboration learning approach for open-world WVAD task using the Evidential Deep Learning (EDL) mechanism based on subjective logic theory. In contrast to conventional classifiers that lack awareness of confidence in their predictions, EDL collects evidence through uncertainty perception. It treats the classification output as a point estimate of the classification distribution, placing priors on the distribution across all possible classification outputs. For the OWVAD task, the assessment of confidence and uncertainty regarding unknown anomalies is significantly more effective through evidence collection and the construction of an uncertainty perception framework. Optimization of prediction results based on this approach proves to be more efficient in many fields compared to traditional classifier learning methods. Therefore, we devise a Multimodal Evidential Collaborative Learning (MECL) module for the OWVAD task. By collecting evidence from visual modality, textual modality, and joint modality, we

Algorithm 1: Training Process of Our Method.

Input: Video V and Category Text T
Output: The final global parameters

- 1 Initialize the network parameters Ψ ;
- 2 **Extract video features:** $X^V \leftarrow \text{VisualEncoder}(V)$;
- 3 **Extract text features:** $X^T \leftarrow \text{TextualEncoder}(T)$;
- 4 **for** *training is not converged* **do**
- 5 **Temporal Modeling:** $X^V \leftarrow \text{MTVM}(X^V)$;
- 6 **Generate Anomaly Scores:**
- 7 $X_N^V \leftarrow \text{Classifier}(X^V)$;
- 8 **Get Visual Attention:**
- 9 $X_N^{V-\text{attn}} \leftarrow \text{Product}(X_N^V, X^V)$;
- 10 **Multimodal Aggregation:**
- 11 $X_n^a \leftarrow \text{VLA}(X_N^{V-\text{attn}}, X^T)$;
- 12 **Aggregation mapping:**
- 13 $X_n^A \leftarrow \text{Product}(X^V, X_n^a)$;
- 14 **Get Multimodal Evidence by Eq.(6)** ;
- 15 **Calculate Joint-Modality Loss:** $\mathcal{L}_{jmd} \leftarrow \text{Eq.(11)}$;
- 16 **Calculate BCE Loss:** $\mathcal{L}_{bce} \leftarrow \text{Top-K}(X_N^V)$;
- 17 **Calculate NCE Loss:** $\mathcal{L}_{nce} \leftarrow \text{MIL}(X_n^A)$;
- 18 **Calculate Total Loss:**
- 19 $L \leftarrow \text{Addition}(\mathcal{L}_{jmd}, \mathcal{L}_{bce}, \mathcal{L}_{nce})$;
- 20 **Update parameters with optimizer** ;
- 21 **end**

Algorithm 2: Inference Process of Our Method.

Input: Video V and Category Text T
Output: The AUC and AP

- 1 **Extract video features:** $X^V \leftarrow \text{VisualEncoder}(V)$
- 2 **Obtain text features:** $X^T \leftarrow \text{Learnable Prompt}(T)$
- 3 **for** *training is not converged* **do**
- 4 **Temporal Modeling:** $X^V \leftarrow \text{MTVM}(X^V)$
- 5 **Generate Anomaly Scores:**
- 6 $X_N^V \leftarrow \text{Classifier}(X^V)$
- 7 **Get Visual Attention:**
- 8 $X_N^{V-\text{attn}} \leftarrow \text{Product}(X_N^V, X^V)$
- 9 **Multimodal Aggregation:**
- 10 $X_n^a \leftarrow \text{VLA}(X_N^{V-\text{attn}}, X^T)$
- 11 **Aggregation mapping:**
- 12 $X_n^A \leftarrow \text{Product}(X^V, X_n^a)$
- 13 **Generate Anomaly Scores:**
- 14 $X_N^T \leftarrow \text{Re-Dimension}(X_n^A)$
- 15 **Calculate AUC and AP:**
- 16 $\text{MetricsFunction}(X_N^V, X_N^T)$
- 17 **end**

optimize prediction results based on the confidence and uncertainty associated with different modalities. For the features of modality $m \in (\text{vision}, \text{language}, \text{aggregated})$, $(x_n^m)_{n=1}^T$, the evidence of the c -th anomaly category is a scalar and can be calculated by:

$$e_{n,c}^m = g(f_c(x_n^m; \theta)), \quad (6)$$

where $f_c(\cdot)$ is the DNN evidential collector parameterized by θ for c -th category. In order to keep the collected evidence

non-negative, the evidence function $g(\cdot)$ usually consists of RELU, SoftPlus or Exp function. For simplicity, we simplify the process of constructing the distribution of evidence for each modality as follow:

$$\text{Distribution}(\mathbf{x}_c; \alpha_c) = \frac{1}{B(\alpha_c)} \prod_{c=1}^C x_c^{\alpha_c-1}, \quad (7)$$

$$B(\alpha_c) = \frac{1}{\Gamma(\sum_{c=1}^C \alpha_c)} \prod_{c=1}^C \Gamma(\alpha_c), \quad (8)$$

where x_c the predicted results of model, $\alpha_c = e_{n,c}^m + 1$, and Γ is the Gama function. Based on this we can obtain the confidence and uncertainty as follow:

$$p_c = \frac{\alpha_c}{\sum_{c=1}^C \alpha_c}, u_c = \frac{c}{\sum_{c=1}^C \alpha_c}. \quad (9)$$

To make the evidence for different modalities effective for joint learning, we jointly optimize using the prediction uncertainty of the different modalities as a correction factor. At first, to prioritise the model to focus on the more reliable modality, we calculate $\{u_c^{uni}, p_c^{uni}\}$ as follow, which represents the confidence of category c in the most representative modality:

$$\{u_c^{uni}, p_c^{uni}\} = \delta(c) \{u_c^v, p_c^v\} + (1 - \delta(c)) \{u_c^t, p_c^t\}, \quad (10)$$

where $\delta(c) = 1$ if $p_c^v > p_c^t$, otherwise the reverse is true.

Then, our multimodal collaborative learning strategy can be expressed as:

$$\begin{aligned} \mathcal{L}_{jmd} = & \sum_c (1 - \text{mean}(u_c^{agg})) (1 - u_c^{uni}) \\ & * l(s(p_c^{agg}), p_c^{uni}), \end{aligned} \quad (11)$$

where $l(\cdot)$ is a distance metrics function, e.g. L2-norm, and $s(\cdot)$ the gradient truncation operation on the input.

IV. EXPERIMENTS

A. Experimental Settings.

Datasets: We perform experiments on the two most popular anomaly detection datasets: (1) **XD-Violence**. The XD-Violence dataset comprises 3954 training videos annotated at the video level and 800 testing videos annotated at the frame level, encompassing 6 categories of violent anomalies. Additionally, it stands out as one of the most challenging datasets due to the frequent occurrence of camera movement and scene transitions within the videos. (2) **UCF-Crime**. The UCF-Crime dataset is a large-scale collection of real-world videos captured by surveillance cameras. Comprising 1,610 training videos annotated with video-level labels and 290 testing videos annotated with frame-level labels for performance evaluation. These surveillance videos are derived from a variety of complex real-world scenarios, covering 13 classes of abnormal events.

Evaluation Metrics: Following the prior work [30], in order to make the testing set contain previously unseen anomalies, we train the model by altering the distribution of labels in the training set. Specifically, we removed different proportions of

anomaly types from the XD-Violence and UCF-Crime training datasets to simulate the real world with different proportions of unseen anomalies. On this basis, we follow previous works and use frame-level Average Precision (AP) and frame-level Area under the operating Characteristic curve (AUC) as the metrics for XD-Violence and UCF-Crime, respectively, on coarse-grained predictions. For fine-grained predictions, we use the mean Average Precision (mAP) under different intersection over union (IoU) thresholds as the evaluation metric.

Implementation Details: For the setting of unseen anomalies in the training set, we train the model with 1, 2, 3, and 4 categories of seen anomaly videos together with normal videos in the XD-Violence dataset, which has a total of 6 anomaly categories in this dataset. Similarly, we train the model with 1, 3, 6, and 9 categories of seen anomaly videos together with normal videos in the UCF-Crime dataset, which has a total of 13 anomaly categories in this dataset. Specifically, we randomly remove a certain number of training anomaly classes each time and repeat them three times to reduce the evaluation bias of the seen anomaly. For the setting of vision-language pre-trained model, we use the Contrastive Language-Image Pre-training (CLIP) to extract both visual and textual features by their corresponding frozen encoders. For the setting of hyperparameters, we set the fusion weight β as 0.2, the weight of $\mathcal{L}_{jmd}$ as 1×10^{-2} , and the temporal context-aware range m as 3. Our model is implemented with Python 3.8 and Pytorch 1.12, and we employ AdamW [53] optimizer with a learning rate of 1×10^{-5} for optimization. All experiments are conducted on a single RTX 3090 GPU.

B. Comparison With State-of-The-Arts

Comparison on XD-Violence: As shown in Table I, our proposed method outperforms previous weakly supervised methods in terms of AP metrics under different conditions of seen anomaly categories. Specifically, compared to the SOTA method VadCLIP [46], our method achieves superior performance under different seen anomaly category conditions, achieving performance gains of 0.35%, 1.39%, 2.03%, and 1.86%, respectively. Compared to other methods, our method also achieves varying degrees of performance gain under different open set conditions. This reflects the superiority of our approach in the open world. As shown in Table III, compared to the performance of other methods under closed-set conditions, our method achieves comparable performance even in the absence of partial seen anomaly data, which reflects its robustness in the face of real-world complexity.

Comparison on UCF-Crime: As shown in Table II, our proposed method outperforms previous weakly supervised methods in terms of AUC metrics under different conditions of seen anomaly categories. Specifically, compared to the SOTA method VadCLIP [46], our method achieves superior performance under different seen anomaly category conditions, achieving performance gains of 0.95%, 0.1%, 0.48%, and 0.34%, respectively. As shown in Table III, we also achieved comparable performance with VadCLIP under closed-set conditions. Similarly, compared to other methods, our method also achieves varying degrees of

TABLE I
COMPARISONS UNDER DIFFERENT SEEN ANOMALY IN XD-VIOLENCE

Method	Venue	Feature	Number of seen Anomaly in XD-Violence(AP/%)			
			1	2	3	4
Wu et al. [49]	ECCV'20	I3D	40.69	50.34	60.53	67.77
RTFM [33]	ICCV'21	I3D	43.54	50.88	58.52	63.65
Zhu et al. [30]	ECCV'22	I3D	46.65	54.65	64.40	69.61
S3R [50]	ECCV'22	I3D	57.81(± 4.01)	76.05(± 1.52)	76.77(± 3.00)	78.01(± 2.12)
UR-DMU [31]	AAAI'23	I3D	62.80(± 14.70)	78.96(± 1.65)	77.61(± 2.54)	78.64(± 4.31)
CLIP-TSA [17]	ICIP'23	CLIP	68.14(± 8.23)	80.52(± 0.74)	80.79(± 1.68)	80.82(± 3.64)
HyperVD [51]	IVC'24	I3D	68.37(± 6.28)	80.05(± 2.52)	81.33(± 2.39)	82.98(± 1.94)
PEI [52]	IEEE TIP'24	I3D	68.70(± 7.37)	79.52(± 2.64)	80.32(± 2.12)	81.00(± 1.68)
VadCLIP [46]	AAAI'24	CLIP	69.12(± 8.45)	80.27(± 3.62)	80.37(± 3.40)	81.58(± 1.96)
Ours	-	CLIP	69.47(± 8.87)	82.26(± 2.44)	82.40(± 3.5)	83.44(± 3.35)

TABLE II
COMPARISONS UNDER DIFFERENT SEEN ANOMALY IN UCF-CRIME

Method	Venue	Feature	Number of seen Anomaly in UCF-Crime(AUC/%) (AP/%)			
			1	3	6	9
Wu et al. [49]	ECCV'20	I3D	73.22	75.15	78.46	79.96
RTFM [33]	ICCV'21	I3D	75.91	76.98	77.68	79.55
Zhu et al. [30]	ECCV'22	I3D	76.73	77.78	78.82	80.14
S3R [50]	ECCV'22	I3D	74.66(± 0.56)	80.09(± 3.18)	82.07(± 1.92)	83.88(± 1.37)
UR-DMU [31]	AAAI'23	I3D	75.23(± 0.68)	81.09(± 2.98)	83.55(± 1.74)	85.12(± 1.54)
CLIP-TSA [17]	ICIP'23	CLIP	78.34(± 1.97)	83.23(± 0.74)	85.93(± 0.97)	87.01(± 0.28)
PEI [52]	IEEE TIP'24	I3D	76.10(± 2.79)	82.27(± 1.82)	84.68(± 0.27)	85.56(± 0.4)
VadCLIP [46]	AAAI'24	CLIP	77.94(± 2.62)	83.41(± 2.45)	86.07(± 0.93)	87.11(± 0.38)
Ours	-	CLIP	78.89(± 1.97)	83.51(± 2.15)	86.55(± 0.84)	87.45(± 0.44)

TABLE III
COMPARISONS UNDER CLOSE-SET SETTING ON XD-VIOLENCE AND UCF-CRIME DATASETS

Method	Venue	Feature	XD	UCF
MIST [54]	CVPR'21	I3D	N/A	82.30
Wu et al. [49]	ECCV'20	I3D	78.64	82.44
RTFM [33]	ICCV'21	I3D	77.81	84.30
MSL [55]	AAAI'22	I3D	78.28	85.30
S3R [50]	ECCV'22	I3D	80.26	85.99
MGFN [56]	AAAI'23	I3D	79.19	86.98
UR-DMU [31]	AAAI'23	I3D	81.66	86.97
ZS CLIP [19]	ICML'21	CLIP	17.83	53.16
Girdhar [57]	CVPR'23	CLIP	25.36	55.78
HyperVD [51]	IVC'24	I3D	85.67	-
PEI [52]	IEEE TIP'24	I3D	85.59	86.76
LLAVA-1.5 [58]	CVPR'24	VIT	50.26	72.84
LAVAD [59]	CVPR'24	VIT	62.01	80.28
Wu et al. [60]	CVPR'24	CLIP	66.53	86.40
CLIP-TSA [17]	ICIP'23	CLIP	83.52	87.58
VadCLIP [46]	AAAI'24	CLIP	84.45	87.83
Ours	-	CLIP	85.13	87.80

performance gain under different open set conditions. This reflects the superiority of our approach in the open world.

In addition, We find that the standard deviation in XD-Violence is much larger than that in UCF-Crime, implying that the selection of different unseen anomalies in XD-Violence has a greater impact on the model. This is particularly severe when

the number of unseen anomalies is 1. We believe that this is mainly due to differences in the video sources of the dataset. The videos in XD-Violence are mainly derived from movies and web videos, which often contain numerous scene changes and complex action behaviors. In contrast, UCF-Crime's videos are mainly derived from surveillance cameras, which have relatively homogeneous scenes and shooting angles.

Performance in Seen and Unseen Categories: In addition, in order to prove that the proposed method has stronger detection ability for the unseen category, we conduct experiments on the seen and unseen categories separately. Specifically, we set Seen Category Set to {Fighting, Shooting, Riot, Abuse} and Unseen Category Set to {Car Accident, Explosion} on XD Violence dataset, while we set Seen Category Set to {Abuse, Arrest, Arson, Assault, Burglary, Explosion, Fighting, Road Accidents, Robbery} and Unseen Category Set to {Shooting, Shoplifting, Stealing, Vandalism} on UCF-Crime dataset. As shown in Tables IV and V, our proposed method achieves better performance than other SOTA methods in both seen and unseen categories. More importantly, the performance improvement of our method on the unseen categories is significantly greater than that on the seen categories, which proves the effectiveness of our method on the unseen category.

Comparison of different methods on each category: In order to explore in more detail the performance of different methods for specific categories in open scenarios, we further conducted

TABLE IV
COMPARISONS BETWEEN SEEN AND UNSEEN CATEGORY IN XD-VIOLENCE

Method	Base (AP/%)	Novel (AP/%)
Random	21.92	31.78
CLIP-TSA	86.83	42.31
VadCLIP	89.39	45.22
Ours	90.65	48.50

TABLE V
COMPARISONS BETWEEN SEEN AND UNSEEN CATEGORY IN UCF-CRIME.

	Base(AUC/%)	Novel(AUC/%)
Random	50.19	50.04
CLIP-TSA	88.54	70.15
VadCLIP	90.01	70.65
Ours	90.25	71.48

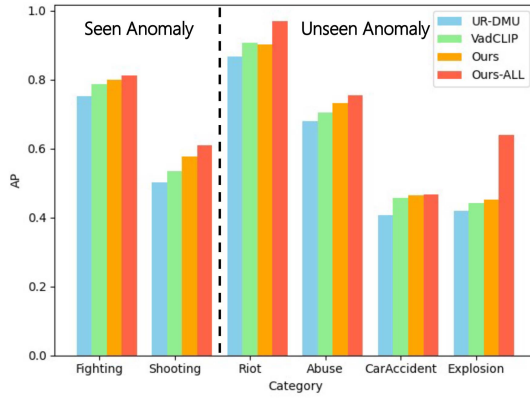


Fig. 3. Comparison of different methods on seen and unseen categories.

experiments for each Base category and Novel category. Specifically, we set Seen Category Set to {Fighting, Shooting} and Unseen Category Set to {Riot, Abuse, Car Accident, Explosion} on XD Violence dataset. As shown in Fig. 3, our method achieves the best performance under open-set conditions in almost all Base and Novel categories, which reflects the effectiveness and superiority of our proposed method. Meanwhile, for the Novel category, the performance under the open-set condition is still quite different from that under the closed-set condition, which means that there is still a lot of scope for improvement in the current open-world video anomaly detection methods.

C. Ablation Study

In this section, ablation experiments are performed under the condition of 4 seen anomaly categories and closed-set to verify the effectiveness of the different modules. Specifically, we investigate the model performance gain of the proposed MTVM and MECL by adding different module combinations in the network. The results of the ablation study are illustrated in Table VI. The MTVM and MECL are independent of each other, and each can achieve better results compared to the baseline. Among them, two modules have achieved performance gains of 0.41% and 1.7% in the 4 seen anomaly category condition, while they have achieved performance gains of 0.43% and 1.67% in the closed set condition. Therefore, we can conclude both MTVM and MECL

TABLE VI
ABLATION STUDY OF MODULES

MTVM	MECL	4 Seen Anomaly	All Seen Anomaly
		AP	AP
×	×	82.39	82.93
✓	×	82.83	83.36
×	✓	84.18	84.60
✓	✓	85.08	85.13

contribute to the performance gain. Moreover, the final result of the ablation study demonstrates that the joint application of proposed modules can further bring a great performance improvement, which proves the effectiveness and rationality of our modules.

It is noteworthy that the proposed MTVM module needs to compute the contextual details of each frame, which imposes a certain computational burden. Specifically, the resulting complexity is mainly related to the video length and feature dimension. Assuming the length of the video is l_v , the feature dimension is d_v , and the number of fused segments is $\mathcal{M}$. Then the computational complexity is $l_v * \mathcal{M} * d_v^2$. Thus, the computational complexity is directly proportional to the length of the video, the longer the video, the more computation is required. It is also proportional to the square of the feature dimension, which means that an increase in the feature dimension leads to a significant increase in the computation. In addition, the number of fused segments $\mathcal{M}$ also affects the computational complexity, with more fused segments implying a more complex computational process. In practical applications, a combination of these factors may need to be considered to balance computational efficiency and model performance. This can ensure that the model can efficiently sense contextual information without putting too much pressure on the computational resources.

Furthermore, the overall performance on close-set is better when the MECL module is used individually. This is due to the fact that the model under the closed-set condition is able to learn wealthier features and patterns from seen anomalous events, thus providing more accurate predictions in most cases. However, the open-set condition brings larger performance gains than the closed-set condition. This indicates that the MECL module is more effective in identifying and distinguishing between normal and unseen abnormal events. This may be associated with the MECL module's ability to better capture the distributional characteristics of the data under open-set conditions, as well as to more accurately evaluate uncertainty.

D. Qualitative Results

In Fig. 4, we show the qualitative results of our proposed method and the baseline (without MTVM and MECL) on unseen and seen anomalies for the XD-Violence dataset. For continuous unseen anomalous events, like riot and shooting, our method produces high confidence in the anomalous region with precise time boundaries, which reflects good detection performance for unseen continuous anomaly. However, the anomalous events are often complex and discontinuous in the real world. As shown in fighting and car accident, it is easy to see that our model still has difficulty in predicting time boundaries of unseen discontinuity

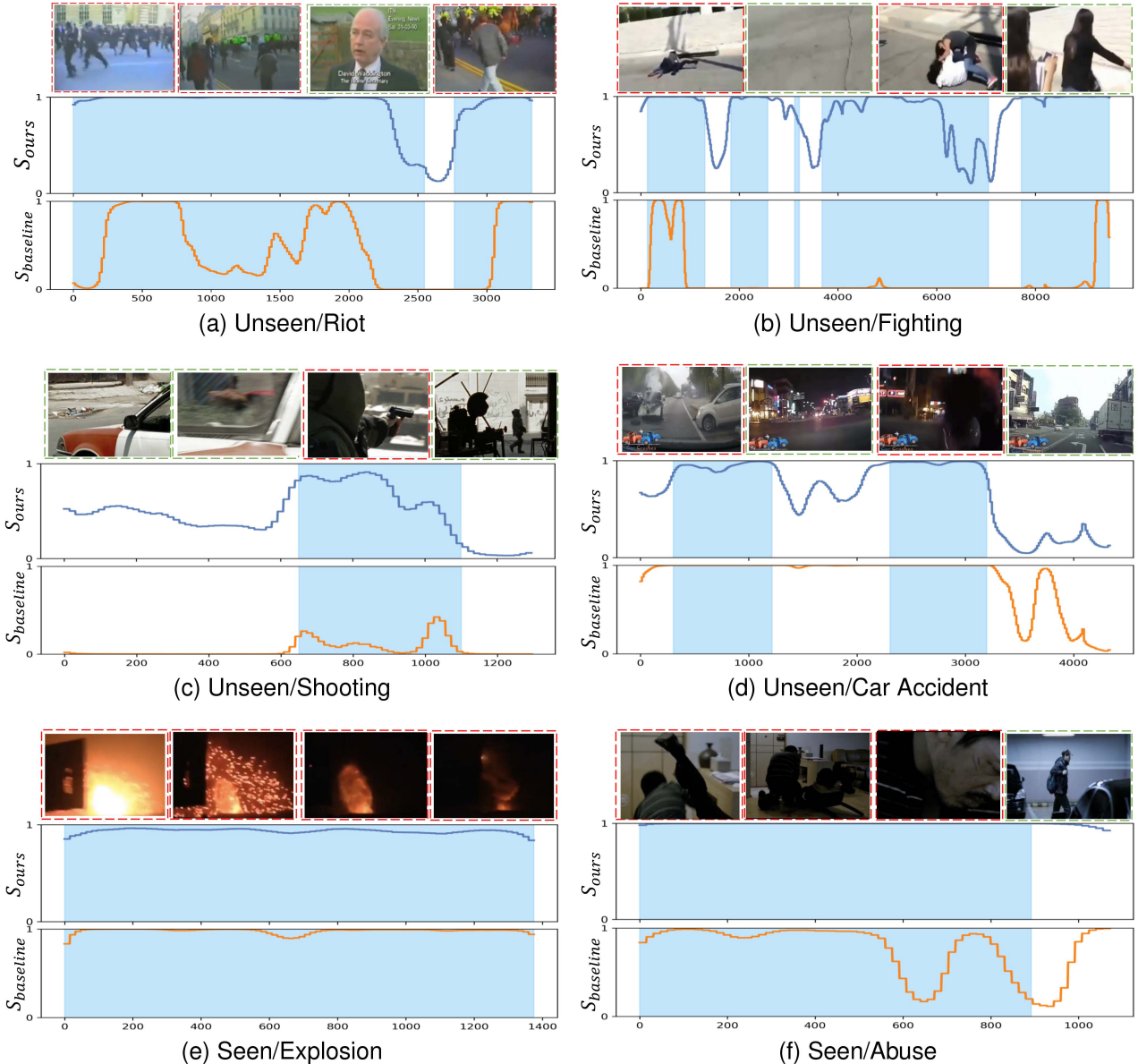


Fig. 4. The qualitative results of our proposed method and the baseline without (MTVM and MECL) on unseen and seen anomalies for the XD-Violence dataset. The light blue range represents abnormal ground truth.

events accurately. And it's worth paying attention to the fact that, for continuous seen anomalous events (like explosion and abuse), our method produces high confidence in the anomalous region with precise time boundaries. These results reflect its similarly positive effect on the categories of seen anomaly.

In conclusion, our method achieves the detection of unseen anomalies under open-world conditions to a certain extent. However, there are still significant challenges in the complex real-world because of the discontinuity and unpredictability of unseen anomalies.

E. Sensitivity Analysis of Parameters

In this section, we discuss the parameters that are critical to model performance. In addition to the hyperparameters

needed for the training process and the learnable parameters in the network, the fusion weight of β plays a significant role in controlling the fusion quality of multi-scale temporal visual modeling features. Therefore, it is essential to explore the appropriate range of β .

As shown in Fig. 5, we investigate the parameter sensitivity regarding the percentage of β for the XD-Violence dataset and UCF-Crime dataset. Specifically, as the weight parameter β increases, the evaluation metrics generally exhibit an initial increase, followed by a decrease, and then another increase.

Regarding the performance in XD-Violence, the optimal model performance is typically achieved when β is approximately 0.1. Regarding the performance in UCF-Crime, the optimal model performance is typically achieved when β is approximately 0.2. Ideally, useful information within the range of

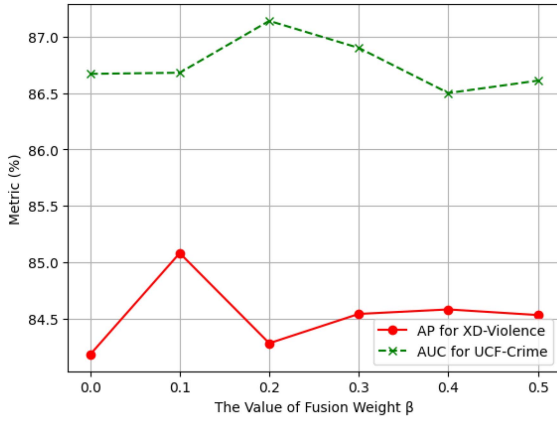


Fig. 5. The changing trend of fusion weight β .

temporal context perception is transferred to the current frame so that it can capture a certain range of temporal context information without affecting the representation of the current frame.

V. CONCLUSION

In this paper, we propose a multi-scale evidential learning method for open-world weakly supervised anomaly detection, which is driven by vision-language pre-trained model. Specifically, we commence by extracting features from both the video and label text using the encoders of the pre-trained model. Specifically, we learn generalized visual and textual feature representation from vision-language pre-trained model and capture useful information in temporal context through multi-scale temporal visual modeling module. We then collect and jointly leverage multimodal evidence to achieve weakly-supervised video anomaly detection in the open world. Extensive experimental results demonstrate the effectiveness of components in our proposed framework, which also show that our method outperforms those existing methods on XD-Violence and UCF-Crime for open-world weakly-supervised video anomaly detection.

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